Project: Worked examples
Title: Shaft: static, fatigue, keys, critical speed
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1. Scope and method

A stepped (optionally hollow) shaft on two bearings carrying radial loads in two planes and a torque with a steady (mean) part and an optional alternating part: bending moment, slope and deflection along it, the fatigue and yield safety factors at the sections the user marks, slopes at the bearings, deflection at the loads, the first lateral critical speed and parallel keys.

WHAT IT CALCULATES	RELATION
Shaft bending moment	the shaft as an S2 beam (pin at A, roller at B) in each plane; $M = \sqrt{(M_y^2 + M_z^2)}$
Stress at a section	$\sigma_a = K_f \cdot 32M_a \cdot d/(\pi(d^4 - d_i^4))$, $\tau_m = K_{fs} \cdot 16T_m \cdot d/(\pi(d^4 - d_i^4))$, $\tau_a = K_{fs} \cdot 16T_a \cdot d/(\pi(d^4 - d_i^4))$; $K_f = 1 + q(K_t - 1)$
Endurance limit	$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_{e1}$, S_{e1} (the rotating-beam limit S_e') $= 0.5 \cdot S_{ut}$ ($S_{ut} \le 1400$ MPa, else 700 MPa); $k_a = a \cdot S_{ut}$^b, $k_b = 1.24d^{-0.107}$ or $1.51d^{-0.157}$, $k_e = 1 - 0.08 \cdot z_R$
Fatigue (DE-Goodman, bending fully reversed, torque $T_m \pm T_a$)	$1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3} \cdot \tau_m/S_{ut}$; yield $n_y = S_y/\sqrt{((\sigma_a)^2 + 3\tau_{pk}^2)}$, τ_{pk} from the larger of $T_m + T_a$ and the peak torque T_{pk} when given
Critical speed	Rayleigh $\omega = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$, y signed (an overhung mass lifts the span, which lowers the quotient); Dunkerley $1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (lower bound, used for the ratio to the operating speed)
Weight loads (with w_{masses})	$F_{y,i} = -m_i \cdot g$ at each node, m_i the carried mass plus the shaft's own mass split to the node; without it the loads table is the whole load
Parallel key, on the peak torque T_k = the larger of $T_m + T_a$ and T_{pk}	$\tau = 2T_k/(d \cdot w \cdot L)$, $\sigma = 4T_k/(d \cdot h \cdot L)$; required length $L = \max(2T_k \cdot n_{target}/(d \cdot w \cdot 0.577 \cdot S_{y,key}), 4T_k \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$, $S_{y,min}$ the lowest of the key, the shaft and (when entered) the hub; a shoulder is checked on both sides and the lower n_f governs

Basis: Fatigue, shaft stresses, deflection, critical speeds, keys[1](10th ed., Ch. 6 (§6-9, Table 6-2 Marin surface factors), Ch. 7 (§7-4 to §7-7)); Shaft design[2](6th ed., Ch. 12).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments	Shaft segments (left to right)	5 row(s)		drawing: GBX-SH-01 rev C	any
x_A	Bearing A position	20	mm	drawing: GBX-SH-01 rev C	0 ... 3000 mm
x_B	Bearing B position	280	mm	drawing: GBX-SH-01 rev C	1 ... 3000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Radial loads and masses (a carried mass adds no weight unless w_{masses} is on: enter its weight as a force in F_y)	1 row(s)		spec: Gear mesh forces from the gear calculation	any
w_{masses}	Add the weight of the carried masses and of the shaft itself as downward vertical loads (off: the loads table is the whole load; the masses then enter the critical speed only)	no		default	any
T	Torque transmitted, steady or mean part T_m	0.3	kN·m	spec: Motor rated torque × service factor (E1)	0 … 50 kN·m
x_{T1}	Torque enters at	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
x_{T2}	Torque leaves at (blank: the shaft length, so every section beyond x_{T1} carries the torque)	330	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
n	Operating speed	1450	rpm	spec: Motor nameplate	0 … 30000 rpm
sections	Sections to check	4 row(s)		assumption: Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit.	any
S_{ut}	Ultimate tensile strength	700	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	300 … 2000 MPa
S_y	Yield strength	500	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	150 … 1900 MPa
E	Young's modulus	207	GPa	assumption: Steel.	50 … 250 GPa
ρ	Density (shaft self-mass in the critical speed)	7850	kg/m³	assumption: Steel.	0 … 20000 kg/m³
finish	Surface finish	machined or cold-drawn		drawing: GBX-SH-01 rev C	any
R	Reliability	99	%	assumption: Design reliability.	50 … 99.9 %
k_d	Temperature factor k_d	1		assumption: Below 250 °C.	0.5 … 1.1
k_f	Miscellaneous factor k_f	1		assumption: No other effects.	0.3 … 1.5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_{target}	Target safety factor	1.5		spec: Company design standard	1.2 … 4
θ_{max}	Slope limit at the bearings	0.001	rad	assumption: Deep-groove ball bearing slope limit.	0.00001 … 0.1 rad
y_{max}	Deflection limit at the loads	0.13	mm	assumption: Spur gear deflection limit.	0.001 … 10 mm
x_{key}	Key position	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
w	Key width	14	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
h	Key height	9	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
L_{key}	Key length	40	mm	drawing: GBX-SH-01 rev C	1 … 500 mm
$S_{y,key}$	Key yield strength	400	MPa	spec: C45 key steel	150 … 1500 MPa

TABLE ROW	VALUES
segments[1]	x_end = 40 mm; d = 35 mm; d_i = 0 mm
segments[2]	x_end = 120 mm; d = 40 mm; d_i = 0 mm
segments[3]	x_end = 180 mm; d = 45 mm; d_i = 0 mm
segments[4]	x_end = 260 mm; d = 40 mm; d_i = 0 mm
segments[5]	x_end = 340 mm; d = 35 mm; d_i = 0 mm
loads[1]	x = 150 mm; F_y = -2.2 kN; F_z = 6 kN; m = 4 kg
sections[1]	x = 120 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[2]	x = 150 mm; K_t = 2.14; K_ts = 3; q = 0.8; q_s = 0.9
sections[3]	x = 180 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[4]	x = 260 mm; K_t = 1.6; K_ts = 1.4; q = 0.8; q_s = 0.9

3. Assumptions

A1. Linear elastic, two simple bearing supports (pin at A, roller at B); bending fully reversed by rotation; torque a steady part T_m plus an alternating part T_a about it (fully alternating when $T_m = 0$), both acting between the torque-in and torque-out positions. (Shigley 10th ed. §7-4.)
A2. No torsional vibration; only the first lateral mode, by Rayleigh with Dunkerley as a lower bound. (Shigley 10th ed. §7-6.)
A3. K_t and K_{ts} come from user records or published fit equations; no stress-concentration chart is reproduced. (No stress-concentration chart is shipped; the factors are entered or derived.)
A4. At a shoulder both sides are checked, each on its own diameter and bore, and the lower fatigue safety factor governs; $k_c = 1$ because the stresses are combined by von Mises. (Shigley 10th ed. §6-9, §7-4.)
A5. The loads table is the whole load: a carried mass adds no weight to the reactions, the moment or the deflection, and neither does the shaft's own weight, unless w_{masses} is on (which adds m·g at the nodes in the vertical plane). (Stated limitation; w_{masses} adds the weights.)

A6. Loads, sections and steps closer together than 1/200 of the shaft length share one node. (Keeps the beam model well conditioned.)
A7. sections (Sections to check) is taken as 4 row(s). (Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_{ts} 3.0); q from the Neuber fit.)
A8. E (Young's modulus) is taken as 207 GPa. (Steel.)
A9. ρ (Density (shaft self-mass in the critical speed)) is taken as 7850 kg/m^3. (Steel.)
A10. R (Reliability) is taken as 99 %. (Design reliability.)
A11. k_d (Temperature factor k_d) is taken as 1. (Below 250 °C.)
A12. k_f (Miscellaneous factor k_f) is taken as 1. (No other effects.)
A13. θ_{max} (Slope limit at the bearings) is taken as 0.001 rad. (Deep-groove ball bearing slope limit.)
A14. y_{max} (Deflection limit at the loads) is taken as 0.13 mm. (Spur gear deflection limit.)

4. Working

[18] $R_A = \sqrt{(R_{A,y}^2 + R_{A,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing A

[19] $R_B = \sqrt{(R_{B,y}^2 + R_{B,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing B

[20] d_A = seat diameter at A
$= x = 20$ mm
$= 35$ mm
Shaft diameter at bearing A

[21] d_B = seat diameter at B
$= x = 280$ mm
$= 35$ mm
Shaft diameter at bearing B

[22] M_{max} = max $\sqrt{(M_y^2 + M_z^2)}$ along the shaft
$= 37$ stations
$= 0.41539$ kN·m
Largest bending moment

[23] θ_A = slope at A (two planes)
$= x = 20$ mm
$= 0.0008961$ rad
Slope at bearing A

[24] θ_B = slope at B (two planes)
$= x = 280$ mm
$= 0.0008961$ rad
Slope at bearing B

[25] y_{load} = largest deflection at a load
$= 0.071775$ mm at 150
$= 0.071775$ mm
Deflection at the loads

[27] $k_a = a \cdot S_{ut}^b$ [1](10th ed. §6-9, Table 6-2)
$= 4.51 \times 700^{-0.265}$ (machined or cold-drawn)
$= 0.79474$
Surface factor

[28] $k_e = 1 - 0.08 \cdot z_R$ [1](10th ed. §6-9)
$= 1 - 0.08 \times 2.3263$
$= 0.81389$
Reliability factor

[29] $S_{e1} = 0.5 \cdot S_{ut}$ (S_e')
= 700 MPa
= 350 MPa
Rotating-beam endurance limit
[30] X1 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 120 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, N·m, σ_a = 79.334 MPa, τ_m = 0 MPa, n_y = 6.3025 (shoulder: the side with d = 40 mm governs, of 2
= 2.3845
Section at 120 mm: fatigue safety factor (DE-Goodman)
[31] X2 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 150 mm, d = 45 mm, k_b = 0.82514, S_e = 186.8 MPa, K_f = 1.912, K_{fs} = 2.8, M = 415.39 N·m, 300 N·m, σ_a = 88.778 MPa, τ_m = 46.947 MPa, n_y = 4.1532
= 1.6909
Section at 150 mm: fatigue safety factor (DE-Goodman)
[32] X3 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 180 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, 300 N·m, σ_a = 79.334 MPa, τ_m = 34.616 MPa, n_y = 5.0281 (shoulder: the side with d = 40 mm gove of 2)
= 1.9801
Section at 180 mm: fatigue safety factor (DE-Goodman)
[33] X4 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 260 mm, d = 35 mm, k_b = 0.84763, S_e = 191.9 MPa, K_f = 1.48, K_{fs} = 1.36, M = 63.906 N·m, 300 N·m, σ_a = 22.47 MPa, τ_m = 48.465 MPa, n_y = 5.7538 (shoulder: the side with d = 35 mm gover of 2)
= 4.2192
Section at 260 mm: fatigue safety factor (DE-Goodman)
[34] $S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_e$' (governing section)
= k_c = 1
= 186.8 MPa
Endurance limit at the governing section
[35] $n_{f,min}$ = min over sections
= 4 section(s)
= 1.6909
Lowest fatigue safety factor
[36] $n_{y,min}$ = min over sections S_y/σ'_max
= 4 section(s)
= 4.1532
Lowest yield safety factor
[37] $n_c = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$ (Rayleigh, y signed) [1](10th ed. §7-6)
= Σm = 7.2337 kg over 37 nodes
= 38275 rpm
First lateral critical speed (Rayleigh)
[38] $n_{c,D} = 1/\omega^2 = \Sigma\ m_i \cdot \delta_{ii}$ (Dunkerley, lower bound) [1](10th ed. §7-6)
= 37 masses
= 37295 rpm
First critical speed, Dunkerley lower bound
[39] $r_c = n_{c,D}/n$
= 37295 / 1450
= 25.721
Critical-speed ratio (Dunkerley lower bound over operating speed)

[44] $\tau_{key} = 2T/(d \cdot w \cdot L)$ [1](10th ed. §7-7)

$= 2 \times 300\ N \cdot m\ /\ (45 \times 14 \times 40)\ mm^3$

$= 23.81\ MPa$

Key shear stress

[45] $\sigma_{key} = 4T/(d \cdot h \cdot L)$ [1](10th ed. §7-7)

$= 4 \times 300\ N \cdot m\ /\ (45 \times 9 \times 40)\ mm^3$

$= 74.074\ MPa$

Key bearing stress

[47] $L_{key,req} = \max(2T \cdot n_{target}/(d \cdot w \cdot 0.577 S_{y,key}),\ 4T \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$

$= n_{target} = 1.5,\ S_{y,min} = 400\ MPa$ (key 400, shaft 500)

$= 11.111\ mm$

Key length required

[48] $u_{key} = L_{req}/L$

$= 11.111 / 40$

$= 0.27778$

Key utilisation

[49] $u_f = n_{target}/n_{f,min}$

$= 1.5 / 1.6909$

$= 0.88712$

Fatigue utilisation

[50] $u_y = n_{target}/n_{y,min}$

$= 1.5 / 4.1532$

$= 0.36117$

Yield utilisation

[51] $u_{slope} = \max(\theta_A,\ \theta_B)/\theta_{max}$

$= 0.0008961 / 0.001$

$= 0.8961$

Slope utilisation

[52] $u_{defl} = y_{load}/y_{max}$

$= 0.071775 / 0.13\ mm$

$= 0.55212$

Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	3.1953 kN	Reaction at A
R_B	3.1953 kN	Reaction at B
d_A	35 mm	Seat diameter at A
d_B	35 mm	Seat diameter at B
M_{max}	0.41539 kN·m	Largest bending moment
θ_A	0.0008961 rad	Slope at A
θ_B	0.0008961 rad	Slope at B
y_{load}	0.071775 mm	Deflection at the loads
S_e	186.8 MPa	Endurance limit
$n_{f,min}$	1.6909	Lowest fatigue safety factor
$n_{y,min}$	4.1532	Lowest yield safety factor
n_c	38275 rpm	First critical speed (Rayleigh)

SYMBOL	VALUE	DESCRIPTION
$n_{c,D}$	37295 rpm	Dunkerley lower bound
r_c	25.721	Critical-speed ratio
τ_{key}	23.81 MPa	Key shear
σ_{key}	74.074 MPa	Key bearing
$L_{key,req}$	11.111 mm	Key length required
u_{key}	0.27778	Key utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Slope at the bearings against limit	$u_{slope} = 0.8961$	$u_{slope} \le 1$	0.896	
PASS	Fatigue safety factor against target (DE-Goodman)	$u_f = 0.88712$	$u_f \le 1$	0.887	[1](10th ed. §7-4)
PASS	Deflection at the loads against limit	$u_{defl} = 0.55212$	$u_{defl} \le 1$	0.552	
PASS	Yield safety factor against target	$u_y = 0.36117$	$u_y \le 1$	0.361	
PASS	Key length against required	$u_{key} = 0.27778$	$u_{key} \le 1$	0.278	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: S_{ut}, S_y, E (Shaft material); θ_{max}, y_{max} (Slope and deflection limits). Enter each as one so its source, edition and licence travel with it.
WARNING — mass-weight: m: the carried mass (4 kg) adds no weight to the reactions, the moment or the deflection, and neither does the shaft's own weight; switch w_{masses} on to add them, or enter the weights as forces in F_y.
WARNING — torque-span: x_{T2}: the torque is transmitted only between x_{T1} = 150 mm and x_{T2} = 330 mm, so the section(s) at 120 mm carry no torque; set x_{T2} (blank: the shaft length) if the torque is carried there.
WARNING — key-hub: $S_{y,hub}$: no hub strength is entered, so the key bearing is checked against the key and the shaft only; the hub is often the weakest part, enter its yield strength.

8. Verification

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Template mech.shaft@1.3.1 Content hash c2b1a293aa8d95dc Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Mott, Machine Elements in Mechanical Design.